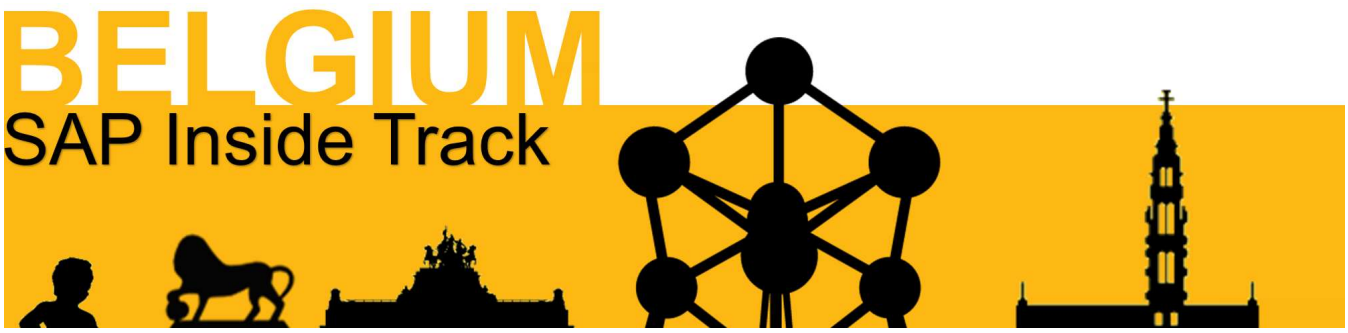
# TypeScript: What's in it for me?

By Jules Decock & Stijn Puttemans

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## This case is presented by...

**Stijn Puttemans**
Full-stack developer
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**Jules Decock**
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# TypeScript: What's in it for me?

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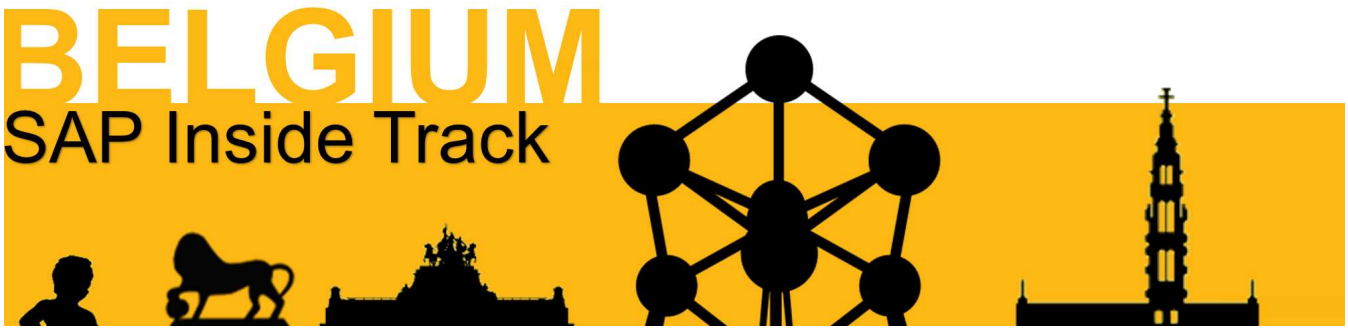




## Agenda

- Some history
- What is Typescript
- Why use Typescript
- Lessons learned
- How to get started

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# Some history

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## JavaScript projects are getting complexer

Some history

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## JavaScript projects are getting complexer

Some history

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## JavaScript projects are getting complexer

Some history

- Hard to navigate
- Prone to bugs
- Hard to keep track of Types
- Refactoring was risky
- ...

- Difficult to maintain
- Hard to scale

TS TypeScript

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# What is TypeScript?

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## TypeScript

What is TypeScript?

A JavaScript Superset

A language building on JavaScript

Browser CANT excute it !!

TS

Transpile

JS

Strongly typed language

Adds new Features and Advantages

Features are compiled to JS “workarounds”. Possible errors are thrown

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## Code example

What is TypeScript?

JS

```
let welcome = "Hello SIT 2023.";

welcome = 2023;

welcome = true;

welcome = "This is awesome!";
```

TS

```
let welcome: string = "Hello SIT 2023.";

welcome = 2023;

welcome = true;

welcome = "This is awesome!";
```

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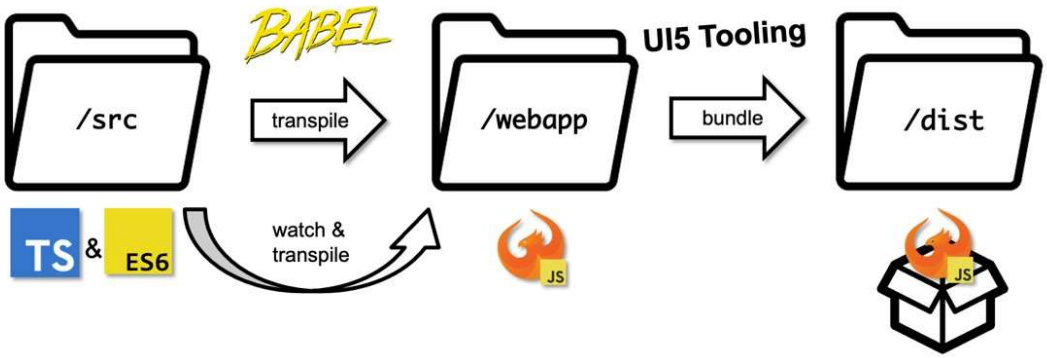
## TypeScript and UI5

What is TypeScript?

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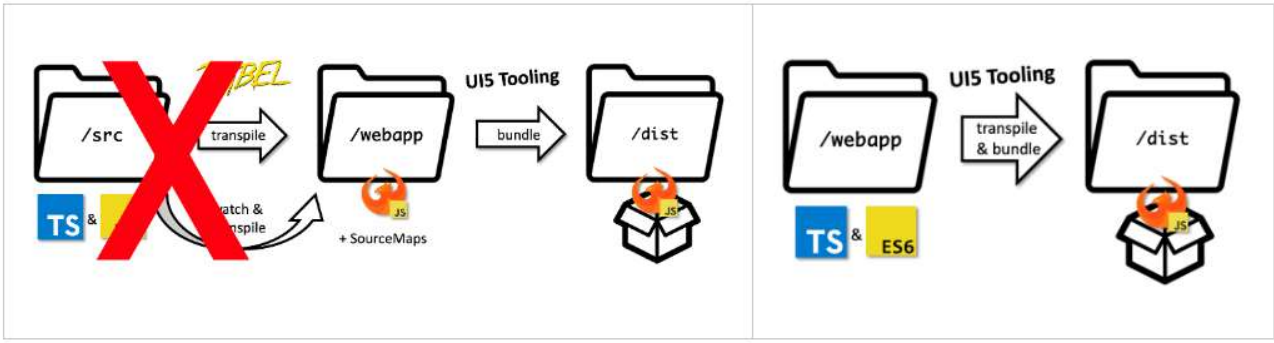
## TypeScript and UI5

What is TypeScript?




**ui5-tooling-transpile** → UI5 Tooling Extension for Transpiling JS/TS

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# Why use TypeScript?

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## Advantages and benefits

Why use TypeScript?

- Good full fledged IDE support
- Early detection of errors
- Extra (OO) features
- ...

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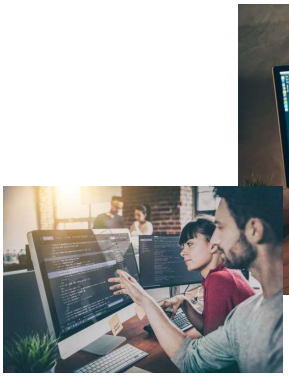
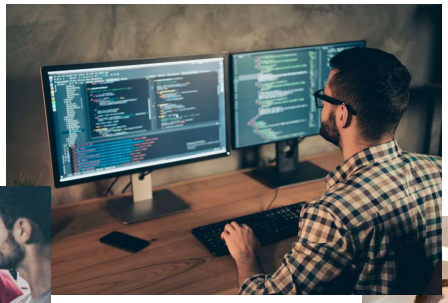
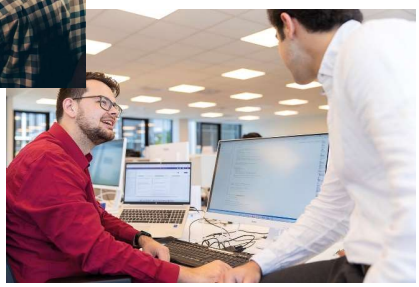
## For developers

Why use TypeScript?







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## Good full-fledged IDE support

Why use TypeScript?

Visual Studio Code

SAP Business Application Studio

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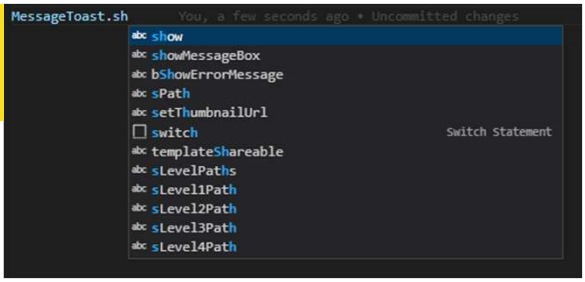
## Good full-fledged IDE support (examples)

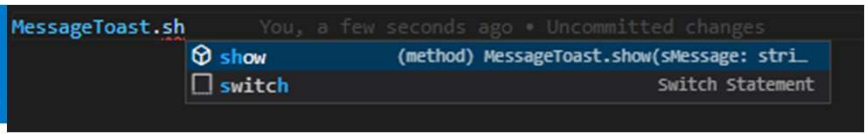
Why use TypeScript?

### Autocompletion and IntelliSense

JS




TS

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## Good full-fledged IDE support (examples)

Why use TypeScript?

### Type Checking and Error Highlighting

JS

```
displayMessage: function(mesg) {
  MessageBox.show(mesg);
},

onDisplayMessage: function() {
  this.displayMessage(4545)
}
```

TS

```
public displayMessage(mesg:string) : void {
    MessageBox.show(mesg);
}

public onDisplayMessage():void {
    this.displayMessage(4545);
}
```

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## Good full-fledged IDE support (examples)

Why use TypeScript?

### Refactoring Support

TS

```
32    public displayMessage(mesg:number) : void {
33        MessageBox.show(mesg);
34    }
35
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL

TS App.controller.ts com.flexso.test1\webapp\controller 1

Argument of type 'string' is not assignable to parameter of type 'number'. ts(2345) [Ln 16, Col 23]

TS BaseController.ts com.flexso.test1\webapp\controller 1

Argument of type 'number' is not assignable to parameter of type 'string'. ts(2345) [Ln 33, Col 19]

TS Main.controller.ts com.flexso.test1\webapp\controller 1

Argument of type 'string' is not assignable to parameter of type 'number'. ts(2345) [Ln 12, Col 23]

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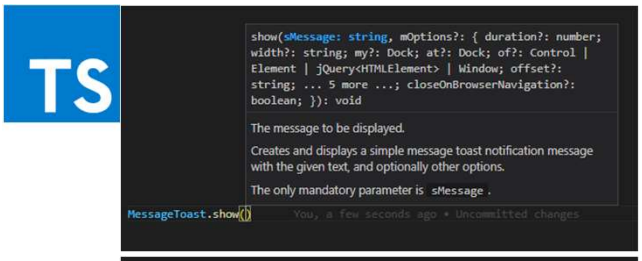
# Good full-fledged IDE support (examples)

Why use TypeScript?

## API Documentation and Navigation



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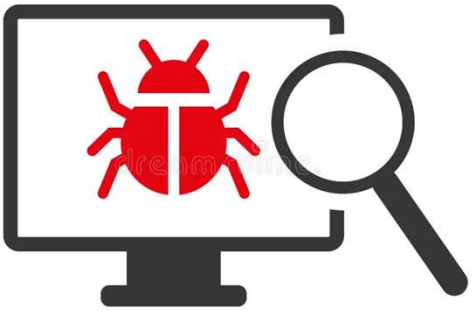
# Early detection of errors

Why use TypeScript?

- Compile-time vs Run-time
- Type Safety
- Interface Contracts
- Refactoring
- (IDE) Support



**False sense of security !!**

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# Extra (OO) features

Why use TypeScript?

## A class based object-oriented programming language structure

- supports features like **classes, interfaces, namespaces, and inheritance**

```typescript
export default class AttachmentState extends BaseState {
    protected service: AttachmentService;
    public data: AttachmentDataType;

    constructor(service: AttachmentService) {
        super();
        this.service = service;
        this.data = {
            documentsList: [],
            documents: [],
            documentsNumber: "?",
            selectedObject: "0000",
            selectedType: "",
            categories: [],
            categoriesForDropDownList: []
        }
        this.getDocumentCategories();
        this.updateModel();
    }

    protected getService(): AttachmentService {
        return this.service;
    }
    protected getData(): AttachmentDataType {
        return this.data;
    }

    private async getDocumentCategories() {
        let cats = await this.getService().getDocumentCategories();
        this.getData().categories = cats;
    }

    public clearCategoriesForDropDownList() {
```

```typescript
import { EmployeeEntity } from "../type/Backend";
import BaseObject from "../core/BaseObject";

/**
 * @namespace com.eliagroup.hr.hrcheckoutv2.model
 */
export class Employee extends BaseObject implements EmployeeEntity {
```

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# Extra (OO) features

Why use TypeScript?

## Supports Native JS modules vs UI5 style AMD modules

- **Import/Export** vs sap.ui.define/require

```typescript
import BaseController from "./BaseController";
import UI5Event from "sap/ui/base/Event";

export type inputParameters = {
    id: string;
    aplzl: string;
    aufpl: string;
};

export type inputMaterialParameters = {
    id: string;
    Number: string;
    ItemNumber: string;
};

/**
 * @namespace com.eliagroup.pm.WoManager.controller
 */
export default class App extends BaseController {
    private currentRouteName: string;
```

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# Extra (OO) features

Why use TypeScript?

## Supports generics

```
type crudFunctions = {
    get: <T>(params?: parameters<T>) => Promise<response<T>>;
    post: <T>(data: T, params?: parameters<T>) => Promise<response<T>>;
    put: <T, R>(data: R, params?: parameters<T>) => Promise<response<T>>;
    delete: <T>(params?: parameters<T>) => Promise<response<T>>;
};
```

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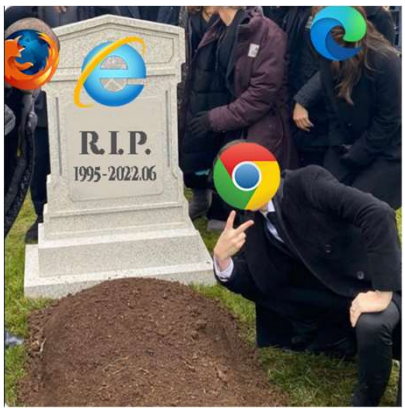
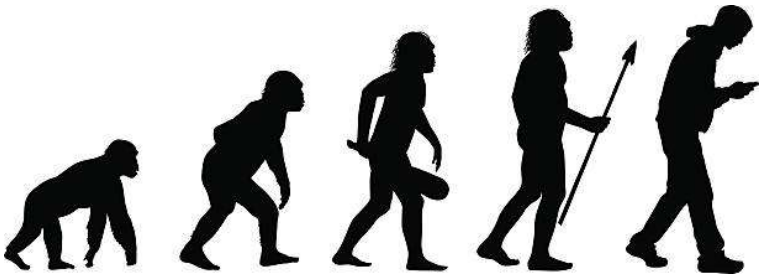
# Future JavaScript features

Why use TypeScript?



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## Self-documenting code

Why use TypeScript?

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## TypeScript is popular and trusted

Why use TypeScript?

**TypeScript is one of the most used languages**

- Big(ger) pool of experts
- Supported by big tech companies like Microsoft
- Community Support
- Availability of Libraries and Tools
- Frequent Updates
- ...

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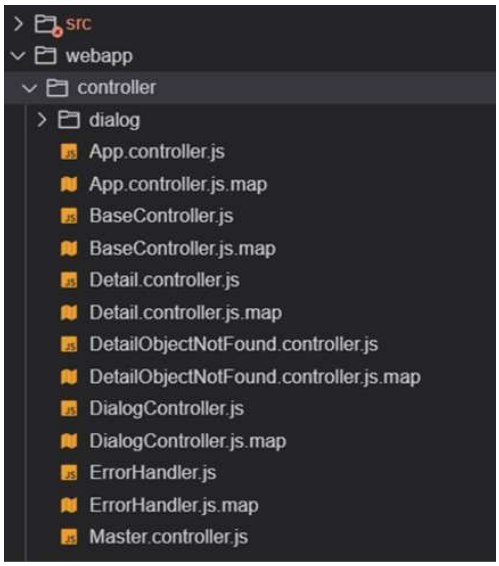
## Debugging with DevTools (F12)

Why use TypeScript?




- Mapping files(.map) are created in the webapp folder
- Chrome's dev tools can consume these files
- Debug TypeScript file instead of the JavaScript file.

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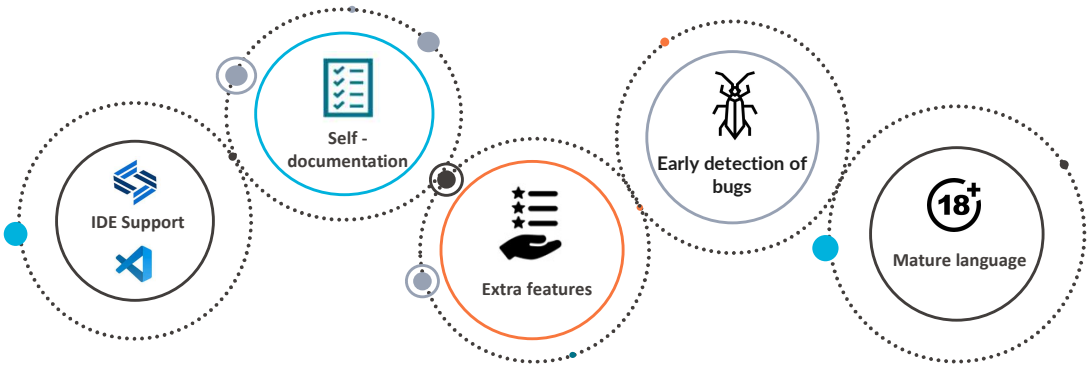
## So why use TypeScript ?

Better overall team performance




**Better overall team performance**

Cleaner code with fewer bugs, faster releases, more confidence in the code quality.

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# Lessons learned

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## Challenges and findings

Lessons learned

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# How to get started

Demo

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# Start using TypeScript

"

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# Q & A

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# Thanks for listening

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# Contact Flexso

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